

长安大学 20_23 —20_24 学年第 2 学期 试题（ A ）卷

课程名称	传感器与测试技术(双语)			考试日期	2024 年 5 月 26 日		共 5 题
学生姓名		学院	工程机械	班级		学号	

Requirement: All the answers should be written on answer sheet. And you can give the answers either in English or in 中文.

一. 填空题（每空 1 分，共 20 分）

1. There are three uncertainty evaluation: standard uncertainty evaluation, _____ and _____.
2. The variable _____ estimates the difference between the mean value of a finite set of measurements and the true value of the measured quantity (the mean of an infinite data set).
3. Periodic signals include _____ signals, _____ signals, _____ signals, and _____ signals.
4. _____ defined as the ratio of change in output of the instrument to change in input or measured variables.
5. The time constant is referred to as the time it takes for the step response to rise to _____ of its steady state value.
6. A sensor can be defined as a device that provides an _____ output with respect to _____.
7. The commonly used temperature error compensation methods in resistance strain sensor system include _____ and _____.
8. The problem of the nonlinearity can be overcome by modifying the single coil system into a _____ reluctance sensor.
9. The flaw detection sensors can detect the crack in some materials. The cracks cause the change of _____ distribution.
10. There are three kinds of variations in capacitance of capacitive sensor _____, _____, _____.
11. The inverse piezoelectric effect occurs when _____ is applied to material, including _____.

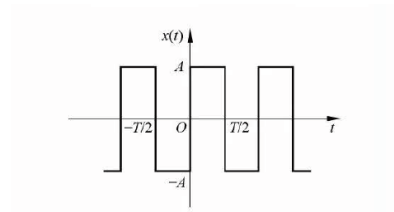
二. 单项选择题（每题 2 分，共 10 分）

- The signal which is the addition of two periodic signals can be regarded a ().
A. Periodic B. Non-periodic
C. Periodic or non-periodic D. Random
- Fourier coefficients a_n and b_n (functions of $n\omega_0 t$) of a periodic signal, a_n is a () function of n or $n\omega_0$.
A. Even B. Odd C. Even or odd D. None
- Given the transfer function $G(s) = \frac{36}{s^2 + 4.2s + 36}$, find ζ . ()
A. $\zeta = 6$ B. $\zeta = 2.1$ C. $\zeta = 0.35$ D. $\zeta = 0.21$
- Which of the following quantities can't be measured using piezoelectric transducers?
()
A. Pressure B. Strain C. Acceleration D. None of them
- Which of the following sensor is commonly used to measure the thickness of a plate?
()
A. Capacitive sensor B. Piezoelectric sensor C. Resistance strain D. all of

1. Measurement can get the true value of the measurand by using good method.
2. A particular problem that can adversely affect the analysis of random errors is the presence of rogue data points(data outliers) in the measurement data.
3. Two signals having periodic components of different frequencies are not correlated.
4. In a distortionless measurement system, the magnitude and phase response must be a constant.
5. Frequency and step-response methods are two main dynamic calibration ways.
6. A 1/4 bridge can achieve temperature compensation. (这个当时记得比较简略, 大概意思)
7. Three kinds of self-generating inductive sensors, including the self-inductance inductance sensor, mutual inductance sensor and eddy current inductance sensors, can realize contact measurement.
8. Skin effect often occurs in a single conductor(i.e., no other conductors are nearby), in which the current density near the surface of the conductor is the largest and the current density increases as the depth increases.

五. 计算题 (3 小题, 共 40 分)

1. A low-pass first-order instrument has a time constant of 20 ms. Find the frequency, in Hz, of the input at which the output will be 93% of the DC output. Find the phase angle at this frequency.
2. Find the Fourier series of the periodic square wave-signal $x(t)$ shown in Fig.



3. 热电偶、最小二乘法

给了一个数据表格，应该是偏最小二乘法的，可能类似中文课本 P25T13

卡西欧最小二乘法教程：<https://zhuanlan.zhihu.com/p/146397890>

13. 铜热电阻与温度之间的关系为 $R_t = R_0(1 + \alpha t)$, 在不同温度下测得的铜热电阻的电阻值见下表。试用最小二乘法原理估计 0°C 时的铜热电阻的电阻值 R_0 和铜热电阻的电阻温度系数 α 。

$t_i/^\circ\text{C}$	15.0	20.0	25.0	30.0	35.0	40.0	50.0
R_{t_i}/Ω	106.42	108.56	110.70	112.84	114.98	117.12	121.40